Technical Document #968: Cybersecurity - Comprehensive Overview

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Vulnerability management identifies, classifies, and remediates security vulnerabilities. Regular scanning and patching are essential components.

Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Incident response plans define how organizations respond to security incidents. They include preparation, detection, containment, and recovery phases.

Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Network segmentation divides networks into smaller segments. It limits the spread of attacks and improves security monitoring.

Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Penetration testing simulates cyber attacks to identify vulnerabilities.
- Social engineering manipulates people into revealing confidential information.
- Incident response plans define how organizations respond to security incidents.